

MAURÍCIO FINAVARO ANICHE

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RESEARCH INTERESTS

Empirical Software Engineering, Mining Software Repositories, Software Testing, Software Maintenance, Software Evolution, and Software Architecture.

EXPERIENCE

Postdoctoral Researcher at TU Delft

2016 - current

Supervisor: Arie van Deursen

Focus: Empirical software engineering, advising of master students, co-teaching software architecture and software testing.

Co-Founder at Alura

2012 - 2015

Focus: Tech leader of the e-learning platform development team; strategic decisions; development of several online courses (37 courses, 169,001 enrollments, median evaluation of 8.84).

Tech Lead and Trainer at Caelum

2010 - 2015

Focus: In-company consulting and training in several topics (Java, software testing, agile methodologies). Total: 1,928 hours, 996 students, 27 companies.

Lecturer at Mackenzie University (Brazil)

2012 - 2015

Focus: Teaching “Software Engineering Topics” (2 editions), and “Data Persistence” (2 editions) for the extension program.

Software Developer at Locaweb

2009 - 2010

Focus: Member of the team responsible for the billing systems; extensive legacy refactoring from Stored Procedures to C#; TDD evangelist.

Software Developer at VeriFone Brazil

2007 - 2009

Focus: Member of the team responsible for the system in charge of the transferring cryptographic keys to POSs in a secure way; embedded development in C; software product lines.

EDUCATION

- **2012 - 2016:** PhD in Computer Science, *University of São Paulo, Brazil*.
- **2009 - 2012:** Master in Computer Science, *University of São Paulo, Brazil*.
- **2004 - 2007:** Bachelor in Computer Science, *Mackenzie University, Brazil*.

PUBLICATIONS

Journal publications

1. **Aniche, M. F.**; Gerosa, M. A. Does test-driven development improve class design? A qualitative study on developers perceptions. *Journal of The Brazilian Computer Society (Online)*, v. 21, p. 15, 2015.
2. **Aniche, M. F.**, Bavota, G., Treude, C., Van Deursen, A., Gerosa, M. A. Code Smells for Model-View-Controller Architectures. *Empirical Software Engineering Journal*, 2017.

Full Papers in Conferences

1. Tómasdóttir, K. F., **Aniche, M. F.**, van Deursen, A. Why and How JavaScript Developers Use Linters. In 32nd IEEE/ACM International Conference on Automated Software Engineering (ASE), 2017.
2. Wieman, R., **Aniche, M. F.**, Lobbezoo, W., Verwer, S., van Deursen, A. An Experience Report on Applying Passive Learning in a Large-Scale Payment Company. In 33rd IEEE International Conference on Software Maintenance and Evolution (ICSME), 2017.
3. Spadini, D., **Aniche, M. F.**, Bruntink, M., Bacchelli, A. To Mock or Not To Mock? An Empirical Study on Mocking Practices. In IEEE 14th International Conference on Mining Software Repositories (MSR), 2017. *Invited to extension*.
4. Van Deursen, A., **Aniche, M. F.**, Aué, J., Slag, R., de Jong, M., Nederlof, A., Bouwers, E. (2017, March). A Collaborative Approach to Teaching Software Architecture. In Proceedings of the 2017 ACM SIGCSE Technical Symposium on Computer Science Education (pp. 591-596). ACM.
5. **Aniche, M. F.**, Treude, C., Zaidman, A., Van Deursen, A., Gerosa, M. A. SATT: Tailoring Code Metric Thresholds for Different Software Architectures. In IEEE 16th International Working Conference on Source Code Analysis and Manipulation (SCAM), 2016.
6. **Aniche, M. F.**, Bavota, G., Treude, C., Van Deursen, A., Gerosa, M. A. A Validated Set of Smells in Model-View-Controller Architectures. In IEEE International Conference on Software Maintenance and Evolution (ICSME), 2016. *Invited to extension*.
7. **Aniche, M. F.**; Gerosa, M. A. What Do The Asserts Tell us About Test-Driven Development?. In: 17th European Conference on Software Maintenance and Reengineering, 2013, Genova. Proceedings of CSMR 2013, 2013.

Full papers in Workshops

1. **Aniche, M. F.**; Silveira, G. Increasing Learning in an Agile Environment: Lessons Learned in an Agile Team. In: Agile Conference, 2011, Salt Lake City. AGILE Conference (AGILE), 2011, 2011. p. 289-295.
2. **Aniche, M. F.**; Gerosa, M. A. Most Common Mistakes in Test-Driven Development Practice: Results from an Online Survey with Developers. In: Third International Conference on Software Testing, Verification and Validation, ICST 2010, 2010, Paris. Workshop on Test driven development (TDD), 2010. p. 469-478.

Pattern papers

1. **Aniche, M. F.**; Guerra, E; Gerosa, M. A. Improving Code Quality on Automated Tests of Web Applications: A Set of Patterns. In: 21st Conference on Pattern Languages of Programs, 2014, Monticello. PLoP 2014, 2014.
2. Guerra, E; **Aniche, M. F.**; Gerosa, M.A.; Yoder, J. Preparing for a Test Driven Development Session. In: 21st Conference on Pattern Languages of Programs, 2014, Monticello. PLoP 2014,

2014.

3. Guerra, E; **Aniche, M. F.**; Gerosa, M. A.; Yoder, J. Test-Driven Development Step Patterns For Designing Objects Dependencies. In: 20th Conference on Pattern Languages of Programs, 2013, Monticello, IL.

Short Papers

1. **Aniche, M. F.** Detection Strategies of Smells in Web Software Development. In: 31st International Conference on Software Maintenance and Evolution (ICSME), Doctoral Symposium, Bremen, 2015.
2. **Aniche, M. F.**; Oliva, G. A. ; Gerosa, M. A. Are All Methods in Your Data Access Objects (DAOs) In The Right Place? A Preliminary Study. In: International Workshop on Managing Technical Debt, 2014.
3. Sokol, F. Z.; **Aniche, M. F.**; Gerosa, M. A.. MetricMiner: Supporting researchers in mining software repositories. In: 2013 IEEE 13th International Working Conference on Source Code Analysis and Manipulation (SCAM), Eindhoven, 2013.

Brazilian Conferences and Workshops

1. Sousa, E., Guerra, E., Oliva, G.A., **Aniche, M. F.** Arquitetura de Aplicações Spring MVC: Uma Análise Baseada no Acoplamento Lógico. In 5th Workshop on Software Visualization, Evolution, and Maintenance (VEM), 2017. *In portuguese.*
2. Wessel, M., Wiese, I., **Aniche, M. F.**, Oliva, G., Gerosa, M. Tweaking Association Rules to Optimize Soware Change Recommendations. In: 31st Brazilian Symposium on Software Engineering (SBES), 2017. *In portuguese.*
3. **Aniche, M. F.**, Treude, C., Gerosa, M. Developers' Perceptions on Object-Oriented Design and System Architecture. In: 30th Brazilian Symposium on Software Engineering (SBES), 2016.
4. **Aniche, M. F.**; Oliva, G. A. ; Gerosa, M. A.. Why Statically Estimate Code Coverage Is So Hard? A Report of Lessons Learned. In: 29o Simpósio Brasileiro de Engenharia de Software (SBES), 2015, Belo Horizonte. 29o Simpósio Brasileiro de Engenharia de Software (SBES), 2015.
5. **Aniche, M. F.**; Gerosa, M. A. Good and Bad Practices in MVC Web Development: Results from a Questionnaire with Professionals. In: III Workshop on Software Visualization, Evolution, and Maintenance, 2015, Belo Horizonte. III Workshop on Software Visualization, Evolution, and Maintenance, 2015. *In portuguese. In portuguese.*
6. França, D. S. ; **Aniche, M. F.**; Guerra, E. How XML Files Evolve? A Study on Its Relation to Source Code. In: 3rd Workshop on Software Visualization, Evolution, and Maintenance (VEM 2015), Belo Horizonte, 2015. *In portuguese.*
7. Sokol, F. Z.; **Aniche, M. F.**; Gerosa, M. A. Does the Act of Refactoring Really Make Code Simpler? A Preliminary Study. In: 4th Brazilian Workshop on Agile Methods (WBMA 2013), 2013, Brasília. Workshop Brasileiro de Métodos Ágeis (WBMA), 2013.
8. **Aniche, M. F.**; Gerosa, M. A. How the Practice of TDD Influences Class Design in Object-Oriented Systems: Patterns of Unit Tests Feedback. In: Simpósio Brasileiro de Engenharia de Software, 2012, Natal. Proceedings of SBES 2013, 2012. *In portuguese.*
9. **Aniche, M. F.**; Ferreira, T.M. ; Gerosa, M. A.. What Concerns Beginner Test-Driven Development Practitioners: A Qualitative Analysis of Opinions in an Agile Conference. In: Workshop Brasileiro de Métodos Ágeis, 2011, Fortaleza. Workshop Brasileiro de Métodos Ágeis, 2011.

10. **Aniche, M. F.**; Gerosa, M. A. A Study on the Impact of Test-Driven Development on Class Design Quality. In: Simpósio Brasileiro de Engenharia de Software, 2010, Salvador. Workshop de Teses e Dissertações em Engenharia de Software, 2010. *In portuguese*.

Book chapters

1. Guerra, E., **Aniche, M. F.** Achieving quality on software design through test-driven development. In: Ivan Mistrik; Richard Soley; Nour Ali; John Grundy; Bedir Tekinerdogan. (Org.). Software Quality Assurance in Large Scale and Complex Software-Intensive Systems. 1ed. Waltham, USA: Morgan Kaufmann – Elsevier, 2015, v. , p. 201-220.

Full Books

1. **Aniche, M. F.** Object-Orientation and SOLID to Ninjas: Projecting flexible classes. 1. ed., 2015. 1,668 copies sold. *In portuguese*.
2. **Aniche, M. F.** Automated Software Testing: A Practical Guide. Casa do Código, 1. ed., 2015. 1,128 copies sold. *In portuguese*.
3. **Aniche, M. F.** Introduction to C programming: The first steps of a developer. Casa do Código, 1. ed., 2015. 407 copies sold. *In portuguese*.
4. **Aniche, M. F.** Test-Driven Development: Test and Design in the Real World. Casa do Código, 1. ed., 2012. 4,235 copies sold. *In portuguese*.

TEACHING

1. Development of 37 online courses to the Alura platform: *VRaptor: More productive web applications*, *Google Chart Tools*, *Design Patterns in Java I*, *Eclipse: Extreme Productivity*, *TDD with Java*, *Mock Objects in Java*, *Selenium*, *C# I*, *C# II*, *C# III*, *Integration tests in Java*, *Java I*, *Design Patterns C# I*, *Tests in C#*, *Spring MVC*, *PHP and MySQL II*, *Testing with Rest Assured*, *JSTL library*, *Refactoring in Java*, *Java II*, *jQuery*, *Design Patterns Java II*, *PHP e MySQL I*, *UML*, *ASP.NET MVC*, *SOLID principles in Java*, *Vagrant*, *Jasmine*, *MySQL I*, *Data Structure*, *Statistics I*, *Statistics II*, *Introduction to C I*, *Introduction to C II*, *Introduction to C III*, *Functional Programming with Clojure*, *Meteor*.
Summary: 37 courses, 169,001 enrollments of which 54,919 completed the course, 36,756 evaluations with an average of 8.84 out of 10 (as of May, 21st 2017).
2. Lecturer at Mackenzie University for 3 years.
Topics: Software Engineering Topics (2 editions) and Data Persistence (2 editions)
3. Online course at Udemy: Learn TDD in 24 hours.
Summary: 466 students from 48 different countries. Overall evaluation: 4,33 out of 5.
4. Co-teaching at TU Delft (2016-2017).
Topics: Software Architecture (2016, 2017, MSc program), Software Testing (2017, Bachelors program).

Tutorials in Conferences

1. **Aniche, M.F.**, Gerosa, M. A. MetricMiner: Mining Source Code Repositories. CBSoft 2015, Belo Horizonte, Brazil. *In portuguese*.
2. **Aniche, M.F.**, Oliva, G. A., Gerosa, M. A., Software Evolution: Tool, Techniques, and Metrics. CBSoft 2011, São Paulo, Brazil. *In portuguese*.

SUPERVISED STUDENTS

1. Jeroen Castelein (TU Delft)
Thesis title: Search-Based Test Data Generation for SQL Queries. MSc program. July/1027.

2. Kristín Fjóla Tómasdóttir (TU Delft)
Thesis title: Why and how developers use linters in JavaScript. MSc program. August/2017.
3. Tom den Braber (TU Delft)
Thesis title: Modeling the Exception Flow in PHP Systems. MSc program. August/2017.
4. Francisco Sokol (USP)
Thesis title: MetricMiner: A tool to ease mining software repositories. Bachelors program. 2012.

ONGOING SUPERVISIONS

Suelen Goulart Carvalho (USP)

Focus: Code smells in Android applications. Following my ICSME 2016 paper. MSc program. Expected conclusion: dec/2017.

Michel Kraaijeveld (TU Delft)

Focus: Automated detection of breaking changes in JavaScript libraries. MSc program. Expected conclusion: aug/2017.

PARTICIPATION IN MSC THESIS COMMITTEES

1. Rick Wieman. *What Does Passive Learning Bring To Adyen?* TU Delft, Mar/2017.
2. Peter Evers. *Finding Relevant Errors in Massive Payment Log Data.* TU Delft, Jan/2017.

SERVICE TO THE COMMUNITY

Program Committes

1. SANER Era Track (International Conference on Software Analysis, Evolution and Reengineering), 2018.
2. SIGCSE (ACM Special Interest Group in Computer Science Education), 2018.
3. SIGCSE (ACM Special Interest Group in Computer Science Education), 2017.
4. CHASE (International Workshop on Cooperative and Human Aspects of Software Engineering), 2017.
5. SCAM (Internal Conference on Source Code and Manipulation), 2017.
6. SANER Era Track (International Conference on Software Analysis, Evolution and Reengineering), 2017.
7. ICSME NIER (International Conference on Software Maintenance), 2017.
8. MSR Challenge track (International Conference on Mining Software Repositories), 2017.
9. SBES Education track (Brazilian Symposium on Software Engineering), 2017.
10. SBES Thesis Workshop (Brazilian Symposium on Software Engineering), 2017.
11. VISSOFT (International Conference on Software Visualization), 2017.
12. WASHES (International Workshop on Social, Human, and Economical Aspects), 2017.
13. VEM (Brazilian Workshop on Software Visualization, Evolution, and Maintenance), 2017.
14. WBMA (Brazilian Workshop on Agile Methods), 2015.
15. PLoP (Pattern Languages of Programming), 2015.
16. WBMA (Brazilian Workshop on Agile Methods), 2014.
17. WBMA (Brazilian Workshop on Agile Methods), 2013.

Chairs

- Experience Report Chair at OSS (International Conference on Open Source Systems), 2014.

External Reviews

1. SIGCSE (ACM Technical Symposium on Computer Science Education), 2017.

2. FSE (International Conference on Foundations of Software Engineering), 2017.
3. ICSME (International Conference on Software Maintenance), 2017.
4. MSR (International Conference on Mining Software Repositories), 2017.
5. ICPC (International Conference on Program Comprehension), 2016.
6. SANER (International Conference on Software Analysis, Evolution and Reengineering), 2016.
7. SBES (Brazilian Symposium on Software Engineering), 2016.

Journal reviews

1. TSE (Transactions on Software Engineering): 2 papers.
2. EMSE (Empirical Software Engineering): 2 papers.
3. JISA (Journal of Internet Services and Applications): 2 papers.
4. IST (Information and Software Technology): 1 paper.

PRESENTATIONS IN INDUSTRY CONFERENCES

1. Code quality in MVC systems. Dutch Spring meetup, 2017.
2. Code quality in MVC systems. BENEVOL (Belgium/Dutch Software Evolution), 2016.
3. Don't you practice TDD? Fatec Zona Leste, São Paulo, 2015. *In portuguese.*
4. How I learned that software testing is important. Mackenzie University, 2015. *In portuguese.*
5. You still don't practice TDD? PUC Poços de Caldas, Brazil, 2015. *In portuguese.*
6. TDD after the mainstream. What now? SERPRO, Brazil, 2015. *In portuguese.*
7. Code metrics: Why do I need you? .NET Architects Day, Brazil, 2015. *In portuguese.*
8. TDD for Managers. Agile Trends, Brazil, 2015. *In portuguese.*
9. Code metrics: Why do I need you? The Developers Conference, Brazil, 2014. *In portuguese.*
10. Is TDD for me? Serra StarTec, Brazil, 2014. *In portuguese.*
11. What is beautiful code? Ticket Tech Day, Brazil, 2013. *In portuguese.*
12. Agile methods: what's folklore and what's real? DevDay 2013, Brazil. *In portuguese.*
13. Code metrics: Why do I need you? Agile Brazil, 2013. *In portuguese.*
14. Code metrics: Why do I need you? Agile Trends Brazil, 2013. *In portuguese.*
15. You still don't practice TDD? Agile Webconf, Brazil, 2013. *In portuguese.*
16. TDD after the mainstream. What now? QCON 2013, Brazil, 2013. *In portuguese.*
17. I measure, you measure, she measures... But what do we measure? Agile Trends, Brazil, 2012. *In portuguese.*
18. Agile methods: what's folklore and what's real? QCON 2012, Brazil. *In portuguese.*
19. Agile methods: what's folklore and what's real? Agile Brazil 2012, Brazil. *In portuguese.*
20. You still don't practice TDD? Agile Tour Poços de Caldas, Brazil, 2012. *In portuguese.*
21. Object-Orientation in the Galaxy. Conexão Java, Brazil, 2011. *In portuguese.*
22. Common mistakes in TDD. Agile Brazil, 2010. *In portuguese.*

23. TDD and its influence on class design. QCON Brazil, 2010. *In portuguese*.
24. Automated Software Testing. .NET Architect Days, Brazil, 2009. *In portuguese*.

OPEN SOURCE DEVELOPMENT

1. RepoDriller. A Java framework that eases mining software repositories.
<http://www.github.com/mauricioaniche/repodriller>. 59 stars and 19 forks on GitHub.
2. SpringLint. Code smells detector and code metrics analysis for Spring MVC applications. The tool was developed during my PhD.
<http://www.github.com/mauricioaniche/springlint>. 40 stars and 1 fork on GitHub.
3. Restfulie.NET. A .NET version of Restfulie (<http://restfulie.caelumobjects.com>), a framework that allows REST level 3 in ASP.NET MVC applications. Now the framework is obsolete as new versions of ASP.NET MVC supports REST natively. 48 stars and 15 forks on Github.